

Experiment	Type	Centaur				Llama-Centaur				Smoltaur				Olmotaur				[binz2025foundation]				Chance	
		0.6B	1.7B	4B	8B	14B	1B	3B	8B	0.1B	0.4B	1.7B	3B	1B	7B	L-70B _p	C _p	C _r	C _p	Cog-p	ln(k)		
Balloon analog risk task [frey2017risk]	Decision	0.07	0.07	0.07	0.06	0.03	0.07	0.07	0.06	0.08	0.07	0.07	0.07	0.07	0.07	0.08	0.07	0.06	0.06	0.09	0.69		
CPC18 [plonsky2018when]	Decision	0.35	0.34	0.34	0.33	0.36	0.35	0.34	0.34	0.42	0.41	0.40	0.34	0.34	0.34	0.41	0.35	0.34	0.66	0.66	0.69		
Columbia card task [frey2017risk]	Decision	0.20	0.20	0.20	0.20	0.20	0.21	0.20	0.20	0.24	0.22	0.21	0.20	0.22	0.20	0.23	0.21	0.19	0.26	0.69	0.69		
Decisions from description [wulff2018sampling]	Decision	0.62	0.63	0.62	0.61	0.60	0.62	0.63	0.60	0.68	0.66	0.64	0.61	0.63	0.62	0.76	0.59	0.53	0.61	0.69	—		
Exponential-symbolic task [garcia2023experiential]	Decision	0.48	0.47	0.46	0.46	0.46	0.47	0.47	0.46	0.57	0.45	0.43	0.47	0.47	0.47	0.70	0.46	0.45	—	—			
Gardening task [flesch2018comparing]	Decision	0.39	0.39	0.38	0.38	0.40	0.39	0.38	0.38	0.57	0.54	0.53	0.38	0.58	0.38	0.50	0.49	0.38	0.91	0.91	0.69		
Go/no-go [enkavi2019large]	Decision	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.08	0.00	0.69		
Intertemporal choice [ruggieri2022globalizability]	Decision	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.47	0.45	0.44	0.44	0.44	0.44	0.73	0.44	0.43	0.66	0.69	0.69		
Multi-attribute DM [hilbig2014generalized]	Decision	0.09	0.08	0.09	0.08	0.08	0.08	0.08	0.08	0.31	0.19	0.16	0.08	0.10	0.09	0.15	0.06	0.06	0.19	0.69	0.69		
Risky choice [krueger2024identifying]	Decision	0.48	0.47	0.44	0.43	0.42	0.49	0.46	0.43	0.72	0.61	0.53	0.45	0.58	0.46	0.65	0.43	0.43	—	—			
choice3k [petersen2021using]	Decision	0.44	0.44	0.43	0.43	0.43	0.44	0.43	0.43	0.49	0.46	0.44	0.44	0.45	0.44	0.53	0.43	0.43	0.66	0.69	0.69		
Multi-task RL [tomov2021multitask]	MDP	0.60	0.59	0.58	0.58	0.57	0.60	0.59	0.57	0.68	0.66	0.63	0.59	0.78	0.59	0.66	0.57	0.57	1.04	1.10	1.10		
Tile-revealing task [kumar2023disentangling]	MDP	0.02	0.01	0.00	0.07	0.00	0.02	0.00	0.00	2.69	1.99	0.04	0.00	0.06	0.09	2.74	2.29	1.87	—	—			
Two-step task [kool2017cost]	MDP	0.54	0.53	0.53	0.53	0.53	0.55	0.53	0.53	0.60	0.60	0.56	0.53	0.57	0.53	0.61	0.53	0.50 [†]	0.60 [†]	0.69	0.69		
Two-step task [kool2016when]	MDP	0.51	0.50	0.49	0.49	0.48	0.52	0.50	0.49	0.59	0.58	0.52	0.50	0.55	0.50	0.61	0.48	0.50 [†]	0.60 [†]	0.69	0.69		
Two-step task [zorowitz2023data]	MDP	0.52	0.51	0.51	0.51	0.51	0.52	0.51	0.51	0.60	0.58	0.52	0.52	0.54	0.51	0.61	0.51	0.50 [†]	0.60 [†]	0.69	0.69		
Virtual subway network [tomov2020discovery]	MDP	1.16	1.15	1.14	1.13	1.12	1.19	1.16	1.13	1.43	1.33	1.30	1.14	1.32	1.15	1.53	1.16	1.13	—	1.61	0.69		
Zoopermarket [ludwig2023human]	MDP	0.54	0.52	0.51	0.50	0.48	0.54	0.52	0.50	0.63	0.61	0.56	0.51	0.61	0.52	0.60	0.50	0.48	0.60	0.69	0.69		
Cond. assoc. learning [collins2014working]	Memory	0.54	0.53	0.52	0.52	0.51	0.55	0.53	0.52	0.74	0.64	0.58	0.53	0.65	0.53	0.64	0.52	0.54	0.86	1.10	1.10		
Digit span [enkavi2019large]	Memory	0.60	0.59	0.58	0.57	0.57	0.60	0.59	0.58	0.71	0.65	0.62	0.59	0.62	0.59	0.66	0.57	0.55	0.94	2.30	2.30		
Episodic long-term memory [popov2023intent]	Memory	1.27	1.05	0.96	0.94	0.90	1.28	1.02	0.92	1.88	1.56	1.37	1.01	1.23	1.05	1.13	0.89	0.87	—	—	0.69		
N-back [enkavi2019large]	Memory	0.42	0.42	0.41	0.41	0.40	0.42	0.42	0.41	0.49	0.47	0.44	0.41	0.44	0.42	0.52	0.40	0.40	0.58	0.69	0.69		
Recall and recognition [cox2018information]	Memory	1.24	1.18	1.13	1.12	1.09	1.22	1.14	1.10	1.69	1.80	1.25	1.15	1.41	1.17	1.38	1.07	1.06	0.69	0.69	0.69		
Recent probes [enkavi2019large]	Memory	0.26	0.26	0.26	0.26	0.26	0.27	0.26	0.26	0.33	0.28	0.27	0.26	0.27	0.26	0.34	0.26	0.39	0.69	0.69	0.69		
Grammar judgement [jansen2021rational]	Misc.	1.47	1.45	1.45	1.44	1.43	1.48	1.45	1.44	1.63	1.54	1.51	1.46	1.51	1.46	1.99	1.44	1.44	1.41	—	—		
Probabilistic reasoning [zhu2020bayesian]	Misc.	2.45	2.40	2.36	2.35	2.34	2.56	2.47	2.43	2.67	2.60	2.50	2.45	2.61	2.47	2.64	2.40	2.37	—	—	—		
Serial reaction time task [wu2023chunking]	Misc.	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.16	0.16	0.16	0.16	0.17	0.19	0.18	0.20	1.39	1.39	1.39		
THINGS odd-one-out [hebart2023things]	Misc.	0.82	0.81	0.80	0.79	0.79	0.82	0.82	0.79	1.08	0.90	0.86	0.80	0.86	0.81	1.14	0.80	0.81	0.83	1.10	1.10		
Changing bandit [xing2023neural]	Bandit	0.30	0.30	0.29	0.29	0.29	0.32	0.31	0.29	0.40	0.38	0.35	0.30	0.39	0.30	0.38	0.45	0.30	0.44	0.69	0.69		
Decisions from experience [wulff2018sampling]	Bandit	0.49	0.49	0.49	0.49	0.48	0.50	0.49	0.48	0.54	0.52	0.50	0.49	0.50	0.49	0.43	0.47	0.37	0.54	—	—		
Drifting four-armed bandit [bahrami2020four]	Bandit	0.73	0.72	0.71	0.71	0.71	0.73	0.72	0.71	0.83	0.78	0.74	0.72	0.74	0.72	0.88	0.71	0.70	0.90	1.39	1.39		
Horizon task [feng2021dynamics]	Bandit	0.23	0.23	0.22	0.22	0.22	0.26	0.24	0.22	0.40	0.34	0.30	0.22	0.35	0.22	0.52	0.40	0.40 [†]	0.36 [†]	0.69	0.69		
Horizon task [sadeghiyeh2020temporal]	Bandit	0.59	0.58	0.58	0.58	0.59	0.58	0.58	0.58	0.62	0.62	0.60	0.58	0.61	0.58	0.52	0.58	0.40 [†]	0.36 [†]	0.69	0.69		
Horizon task [somerville2017charting]	Bandit	0.34	0.34	0.34	0.34	0.33	0.35	0.34	0.34	0.48	0.40	0.38	0.34	0.34	0.34	0.52	0.35	0.40 [†]	0.36 [†]	0.69	0.69		
Horizon task [waltz2020differential]	Bandit	0.15	0.15	0.15	0.15	0.15	0.16	0.16	0.14	0.26	0.21	0.18	0.15	0.19	0.15	0.52	0.15	0.40 [†]	0.36 [†]	0.69	0.69		
Horizon task [wilson2014humans]	Bandit	0.46	0.46	0.45	0.45	0.45	0.47	0.46	0.45	0.54	0.51	0.49	0.45	0.52	0.45	0.52	0.48	0.40 [†]	0.36 [†]	0.69	0.69		
Iowa gambling task [steingrover2015data]	Bandit	0.93	0.93	0.91	0.91	0.91	0.94	0.92	0.91	1.01	0.98	0.94	0.92	0.95	0.92	0.99	0.91	0.89	1.16	1.39	1.39		
Prob. instrumental learning [lefebvre2017behavioural]	Bandit	0.50	0.50	0.49	0.49	0.49	0.51	0.50	0.49	0.54	0.53	0.51	0.50	0.51	0.50	0.54	0.50	0.49	0.50	0.69	0.69		
Spatially correlated MAB [wu2018generalization]	Bandit	2.01	1.96	1.93	1.87	1.86	2.16	2.00	1.93	2.50	2.30	2.16	1.95	2.26	2.06	2.45	1.82	1.83	2.76	3.40	3.40		
Structured bandit [schulz2020finding]	Bandit	0.67	0.66	0.65	0.65	0.64	0.67	0.66	0.65	0.77	0.71	0.68	0.65	0.73	0.66	0.81	0.64	0.64	1.05	2.08	2.08		
Two-armed bandit [gershman2018deconstructing]	Bandit	0.30	0.30	0.29	0.29	0.29	0.30	0.29	0.29	0.37	0.33	0.31	0.29	0.31	0.30	0.38	0.30	0.30	0.42	0.69	0.69		
Aversive learning [wise2019computational]	Sup. learn.	4.71	4.55	4.27	4.24	3.81	4.58	4.80	3.66	5.35	5.12	4.92	3.93	5.44	4.67	5.11	4.13	4.07	—	—	—		
Medin categorization [levering2020revisiting]	Sup. learn.	0.51	0.51	0.50	0.50	0.50	0.52	0.50	0.50	0.56	0.54	0.52	0.50	0.52	0.51	0.58	0.50	0.50	0.53	—	—		
Multiple-cue judgment [collisioo2023numerical]	Sup. learn.	1.17	1.16	1.14	1.14	1.13	1.17	1.15	1.12	1.59	1.49	1.40	1.14	1.67	1.15	1.28	1.14	1.12	1.92	2.20	2.20		
Shepard categorization [badham2017deficits]	Sup. learn.	0.55	0.55	0.54	0.53	0.53	0.56	0.54	0.53	0.62	0.58	0.56	0.54	0.56	0.54	0.58	0.54	0.54	0.61	0.69	0.69		
Weather prediction task [speekenbrink2008learning]	Sup. learn.	0.55	0.56	0.54	0.55	0.55	0.54	0.55	0.54	0.56	0.58	0.56	0.55	0.57	0.53	0.57	0.56	0.55	0.63	0.69	0.69		
Mean		0.69	0.68	0.66	0.66	0.64	0.70	0.68	0.64	0.89	0.82	0.73	0.66	0.77	0.68	0.88	0.71	0.69	0.69	0.98	0.98		

Bold+underline = best; **underline** = second-best (C_p excluded; see below). Lower is better. “—” = not available. All models finetuned at LoRA rank 16, bf16 inference. Reference columns ([binz2025foundation]): L-70B_p = Llama-3.1-70B base, C_r = our reproduced Centaur-70B, C_p = published Centaur-70B, Cog-p = domain-specific cognitive baseline. C_p is included for reference but excluded from best/second-best marking, as it was evaluated under different software conditions. [†]Horizon and Two-step tasks are reported as single merged values by [binz2025foundation]. Rounded to 2 d.p.

Experiment	Type	Qwen3 family										Llama-3.1/3.2 family										[binz2025foundation]										Chance		
		0.6B		1.7B		4B		8B		14B		1B		3B		8B		L-70B _p		C _p		C _r		C _g		C _g		C _g		C _g		ln(<i>k</i>)		
		base	ft	base	ft	base	ft	base	ft	base	ft	base	ft	base	ft	base	ft	base	ft	base	ft	base	ft	base	ft	base	ft	base	ft	base	ft	base	ft	
Balloon analog risk task [frey2017risk]	Decision	0.09	0.07	0.09	0.07	0.08	0.07	0.08	0.06	0.08	0.06	0.10	0.07	0.08	0.07	0.08	0.06	0.08	0.07	0.07	0.06	0.08	0.07	0.06	0.09	0.09	0.09	0.09	0.09	0.09	0.09	0.09	0.69	
CPC18 [plonsky2018when]	Decision	0.41	0.35	0.40	0.34	0.40	0.34	0.40	0.34	0.40	0.34	0.43	0.36	0.41	0.35	0.39	0.34	0.41	0.35	0.34	0.36	0.41	0.35	0.34	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.69	
Columbia card task [frey2017risk]	Decision	0.27	0.20	0.26	0.20	0.26	0.20	0.25	0.20	0.25	0.20	0.30	0.21	0.27	0.20	0.25	0.20	0.23	0.21	0.19	0.26	0.23	0.21	0.19	0.26	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69
Decisions from description [wulf2021sampling]	Decision	1.08	0.62	1.05	0.63	1.01	0.62	0.99	0.61	1.01	0.60	1.55	0.62	1.20	0.63	0.96	0.60	0.76	0.59	0.45	0.76	0.46	0.70	0.49	0.45	—	—	—	—	—	—	—	—	
Exponential-symbolic task [garcia2023experiential]	Decision	0.76	0.48	0.68	0.47	0.67	0.46	0.64	0.46	0.57	0.46	0.78	0.47	0.69	0.47	0.64	0.46	0.70	0.49	0.47	0.69	0.47	0.64	0.46	0.70	0.49	0.45	—	—	—	—	—	—	
Gardening task [flesch2018comparing]	Decision	0.52	0.39	0.52	0.39	0.49	0.38	0.47	0.38	0.45	0.38	0.81	0.40	0.64	0.39	0.48	0.38	0.50	0.49	0.38	0.50	0.49	0.38	0.50	0.49	0.38	0.50	0.49	0.38	0.50	0.49	0.38	0.50	0.69
Go/no-go [enkavi2019large]	Decision	0.01	0.00	0.01	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.01	0.00	0.01	0.00	0.00	0.00	0.01	0.00	0.00	0.01	0.00	0.00	0.01	0.00	0.00	0.01	0.00	0.00	0.01	0.00	0.00	0.00	
Go/no-go [enkavi2019large]	Decision	0.01	0.00	0.01	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.01	0.00	0.01	0.00	0.00	0.00	0.01	0.00	0.00	0.01	0.00	0.00	0.01	0.00	0.00	0.01	0.00	0.00	0.01	0.00	0.00	0.00	
Intertemporal choice [ruggieri2022globalizability]	Decision	0.77	0.44	0.80	0.44	0.71	0.44	0.69	0.44	0.70	0.44	0.85	0.44	0.75	0.44	0.76	0.44	0.73	0.44	0.73	0.44	0.73	0.44	0.73	0.44	0.66	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69
Intertemporal choice [ruggieri2022globalizability]	Decision	0.49	0.09	0.34	0.08	0.20	0.09	0.19	0.08	0.19	0.08	0.63	0.08	0.45	0.08	0.23	0.08	0.15	0.06	0.06	0.19	0.15	0.06	0.06	0.19	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.15
Risky choice [krueger2024identifying]	Decision	0.92	0.48	0.83	0.47	0.79	0.44	0.74	0.43	0.67	0.42	0.99	0.49	0.85	0.46	0.76	0.43	0.65	0.43	0.43	0.43	0.65	0.43	0.43	—	—	—	—	—	—	—	—	—	
Risky choice [krueger2024identifying]	Decision	0.62	0.44	0.60	0.44	0.59	0.43	0.59	0.43	0.55	0.43	0.67	0.44	0.59	0.44	0.55	0.43	0.53	0.43	0.43	0.65	0.43	0.43	0.65	0.43	0.43	0.65	0.43	0.43	0.65	0.43	0.65	0.69	
choices13k [peterson2021using]	Decision	0.62	0.44	0.60	0.44	0.59	0.43	0.59	0.43	0.55	0.43	0.67	0.44	0.59	0.44	0.55	0.43	0.53	0.43	0.43	0.65	0.43	0.43	0.65	0.43	0.43	0.65	0.43	0.43	0.65	0.43	0.65	0.69	
Multi-task RL [tomov2021multitask]	MDP	0.70	0.60	0.72	0.59	0.68	0.58	0.70	0.58	0.68	0.57	0.80	0.60	0.70	0.59	0.66	0.57	0.66	0.57	0.57	0.57	0.66	0.57	0.57	1.04	1.10	1.10	1.10	1.10	1.10	1.10	1.10	1.10	
Title-revealing task [kumar2023disentangling]	MDP	2.95	0.02	2.81	0.01	2.67	0.00	2.73	0.07	2.60	0.00	4.86	0.02	3.98	0.00	2.92	0.00	2.74	2.29	1.87	—	—	—	—	—	—	—	—	—	—	—	—	—	
Two-step task [kool2017cost]	MDP	0.67	0.54	0.64	0.53	0.64	0.53	0.65	0.53	0.63	0.53	0.67	0.55	0.62	0.53	0.61	0.53	0.61	0.53	0.50 [†]	0.60 [†]	0.61	0.53	0.50 [†]	0.60 [†]	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69
Two-step task [kool2016when]	MDP	0.69	0.51	0.64	0.50	0.64	0.49	0.65	0.49	0.63	0.48	0.67	0.52	0.62	0.50	0.61	0.49	0.61	0.48	0.50 [†]	0.60 [†]	0.61	0.48	0.50 [†]	0.60 [†]	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69
Two-step task [zorowitz2023data]	MDP	0.62	0.52	0.59	0.51	0.60	0.51	0.61	0.51	0.59	0.51	0.64	0.52	0.60	0.51	0.58	0.51	0.61	0.51	0.50 [†]	0.60 [†]	0.61	0.51	0.50 [†]	0.60 [†]	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69
Virtual subway network [tomov2020discovery]	MDP	1.68	1.16	1.55	1.15	1.57	1.14	1.55	1.13	1.57	1.12	1.68	1.19	1.60	1.16	1.53	1.13	1.53	1.16	1.13	—	—	—	—	—	—	—	—	—	—	—	—	—	
Zoo-permarket [ludwig2023human]	MDP	0.61	0.54	0.61	0.52	0.60	0.51	0.60	0.50	0.59	0.48	0.64	0.54	0.62	0.52	0.60	0.50	0.60	0.50	0.48	0.60	0.50	0.48	0.60	0.50	0.48	0.60	0.50	0.48	0.60	0.50	0.48	0.60	
Cond. assoc. learning [collins2014working]	Memory	0.76	0.54	0.71	0.53	0.71	0.52	0.69	0.52	0.61	0.51	0.82	0.55	0.73	0.53	0.70	0.52	0.64	0.52	0.54	0.86	0.52	0.54	0.86	0.52	0.54	0.86	0.52	0.54	0.86	0.52	0.54	0.86	1.10
Digit span [enkavi2019large]	Memory	0.94	0.60	0.89	0.59	0.72	0.58	0.70	0.57	0.69	0.57	1.01	0.60	0.84	0.59	0.75	0.58	0.66	0.57	0.55	0.94	0.57	0.55	0.94	0.57	0.55	0.94	0.57	0.55	0.94	0.57	0.55	0.94	2.30
Episodic long-term memory [popov2023intent]	Memory	1.72	1.27	1.45	1.05	1.33	0.96	1.30	0.94	1.18	0.94	1.97	1.28	1.65	1.02	1.43	0.92	1.13	0.89	0.87	—	—	—	—	—	—	—	—	—	—	—	—	—	0.69
N-back [enkavi2019large]	Memory	0.58	0.42	0.58	0.42	0.55	0.41	0.55	0.41	0.53	0.40	0.59	0.42	0.57	0.42	0.57	0.41	0.52	0.40	0.40	0.58	0.40	0.52	0.40	0.40	0.58	0.40	0.58	0.40	0.58	0.40	0.58	0.40	0.69
Recall and recognition [cox2018information]	Memory	1.73	1.24	1.57	1.18	1.44	1.13	1.41	1.12	1.36	1.09	1.87	1.22	1.56	1.14	1.37	1.10	1.38	1.07	1.06	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Recent probes [enkavi2019large]	Memory	0.70	0.26	0.61	0.26	0.39	0.26	0.41	0.26	0.35	0.26	0.74	0.27	0.56	0.26	0.34	0.26	0.34	0.26	0.26	0.39	0.26	0.39	0.26	0.39	0.26	0.39	0.26	0.39	0.26	0.39	0.26	0.39	0.69
Grammar judgement [jansen2021rational]	Misc.	2.12	1.47	2.02	1.45	2.00	1.45	1.89	1.44	1.91	1.43	2.33	1.48	2.16	1.45	2.12	1.44	1.99	1.44	1.44	1.41	1.99	1.44	1.44	—	—	—	—	—	—	—	—	—	—
Probabilistic reasoning [zhu2020bayesian]	Misc.	2.75	2.45	2.69	2.40	2.61	2.36	2.57	2.35	2.58	2.34	2.89	2.56	2.80	2.47	2.71	2.43	2.64	2.40	2.37	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Serial reaction time task [wu2023chunking]	Misc.	0.21	0.17	0.19	0.17	0.19	0.17	0.19	0.17	0.19	0.17	0.31	0.17	0.23	0.17	0.20	0.17	0.19	0.18	0.17	0.20	0.18	0.17	0.20	0.18	0.17	0.20	0.18	0.17	0.20	0.18	0.17	0.20	1.39
THINGS odd-one-out [hebart2023things]	Misc.	1.21	0.82	1.21	0.81	1.17	0.80	1.15	0.79	1.13	0.79	1.31	0.82	1.18	0.82	1.16	0.79	1.14	0.80	0.81	0.83	1.14	0.80	0.81	0.83	1.14	0.80	0.81	0.83	1.14	0.80	0.81	0.83	1.10
Changing bandit [xiong2023neural]	Bandit	0.42	0.30	0.38	0.30	0.36	0.29	0.35	0.29	0.34	0.29	1.34	0.32	0.90	0.31	0.39	0.29	0.38	0.45	0.30	0.44	0.43	0.47	0.37	0.54	—	—	—	—	—	—	—	—	0.69
Decisions from experience [wulf2021sampling]	Bandit	0.74	0.40	0.66	0.49	0.69	0.49	0.78	0.49	0.61	0.48	0.81	0.50	0.71	0.49	0.65	0.48	0.43	0.47	0.37	0.54	0.43	0.47	0.37	0.54	—	—	—	—	—	—	—	—	—
Drifting four-armed bandit [bahrami2020four]	Bandit	0.98	0.73	0.95	0.72	0.92	0.71	0.91	0.71	0.91	0.71	0.95	0.73	0.94	0.72	0.90	0.71	0.88	0.71	0.70	0.90	0.88	0.71	0.70	0.90	1.39	1.39	1.39	1.39	1.39	1.39	1.39	1.39	1.39
Horizon task [feng2021dynamics]	Bandit	0.47	0.23	0.43	0.23	0.39	0.22	0.36	0.22	0.34	0.22	0.81	0.26	0.67	0.24																			

Task type	Qwentaur					Llama-Centaur			Smoltaur				Olmotaur		Base			[binz2025foundation]		
	0.6B	1.7B	4B	8B	14B	1B	3B	8B	0.1B	0.4B	1.7B	3B	1B	7B	Q-8B	Q-14B	L-8B	70B _p	70B _r	Cog _p
Decision (11)	0.32	0.32	0.32	0.31	<u>0.31</u>	0.32	0.32	<u>0.31</u>	0.40	0.37	0.35	0.32	0.36	0.32	0.46	0.44	0.46	0.30	0.32	0.46
MDP (7)	0.56	0.55	0.54	0.54	<u>0.53</u>	0.56	0.54	<u>0.53</u>	1.03	0.91	0.59	0.54	0.63	0.56	1.07	1.04	1.07	0.79	0.86	0.69
Bandit (13)	0.59	0.59	0.58	<u>0.57</u>	<u>0.57</u>	0.61	0.59	0.58	0.71	0.66	0.63	0.58	0.65	0.59	0.74	0.73	0.76	0.58	0.60	0.74
Memory (6)	0.72	0.67	0.65	0.64	<u>0.62</u>	0.72	0.66	0.63	0.97	0.85	0.76	0.66	0.77	0.67	0.84	0.78	0.86	0.61	<u>0.62</u>	0.69
Misc. (4)	1.23	1.21	1.19	1.19	<u>1.18</u>	1.26	1.23	1.21	1.39	1.30	1.25	1.22	1.28	1.23	1.45	1.45	1.55	1.20	1.20	<u>0.81</u>
Sup. learn. (5)	1.50	1.47	1.40	1.39	1.30	1.48	1.51	<u>1.27</u>	1.74	1.66	1.59	1.33	1.75	1.48	1.59	1.55	1.66	1.36	1.37	<u>0.92</u>
Mean (46)	0.69	0.68	0.66	0.66	<u>0.64</u>	0.70	0.68	<u>0.64</u>	0.89	0.82	0.73	0.66	0.77	0.68	0.89	0.87	0.92	0.69	0.71	0.69
<0.20 0.20–0.40 0.40–0.60 0.60–0.80 0.80–1.00 >1.00																				

Table 3: Mean NLL by task type for finetuned models (bf16). Finetuned families: **Qwentaur**, **Llama-Centaur**, **Smoltaur**, **Olmotaur**. Base columns: Q-8B/Q-14B (Qwen3), L-8B (Llama-3.1). Subscript _r denotes our reproducing evaluation of the original Centaur model distributed by [binz2025foundation], evaluated under identical python library and CUDA versions as our small foundation models for fair comparison. Subscript _p denotes values published by [binz2025foundation]. 70B_p is shown for reference but excluded from best/second-best marking, as it was evaluated under different software conditions. Cell colour encodes performance; **bold+underline** marks the best model, underline the second-best. Full per-experiment results in Appendix ??.

Task type	Qwentaur					Llama-Centaur			Smoltaur				Olmotaur		Base			[binz2025foundation]		
	0.6B	1.7B	4B	8B	14B	1B	3B	8B	0.1B	0.4B	1.7B	3B	1B	7B	Q-8B	Q-14B	L-8B	70B _p	70B _r	Cog _p
Decision (9)	0.29	0.29	0.29	0.28	<u>0.28</u>	0.29	0.29	<u>0.28</u>	0.36	0.33	0.32	0.28	0.33	0.29	0.41	0.40	0.41	0.27	0.29	0.46
MDP (5)	0.54	0.53	0.52	0.52	<u>0.51</u>	0.54	0.53	0.52	0.62	0.60	0.56	0.53	0.61	0.53	0.64	0.62	0.61	0.51	<u>0.52</u>	0.69
Bandit (13)	0.59	0.59	0.58	<u>0.57</u>	<u>0.57</u>	0.61	0.59	0.58	0.71	0.66	0.63	0.58	0.65	0.59	0.74	0.73	0.76	0.58	0.60	0.74
Memory (4)	0.45	0.45	0.44	0.44	<u>0.44</u>	0.46	0.45	0.44	0.57	0.51	0.48	0.45	0.49	0.45	0.59	0.54	0.59	0.44	<u>0.44</u>	0.69
Misc. (3)	0.82	0.81	0.80	0.80	<u>0.80</u>	0.83	0.81	<u>0.80</u>	0.96	0.87	0.84	0.81	0.84	0.81	1.08	1.08	1.16	0.80	0.81	0.81
Sup. learn. (4)	0.69	0.70	0.68	0.68	<u>0.68</u>	0.70	0.68	<u>0.68</u>	0.84	0.79	0.76	0.68	0.83	0.68	0.77	0.75	0.77	0.68	0.68	0.92
Mean (38)	0.53	0.52	0.52	<u>0.51</u>	<u>0.51</u>	0.54	0.52	0.51	0.64	0.59	0.56	0.52	0.59	0.52	0.66	0.65	0.67	0.51	0.52	0.69
<0.20 0.20–0.40 0.40–0.60 0.60–0.80 0.80–1.00 >1.00																				

Table 4: Mean NLL by task type for finetuned models (bf16), restricted to the 38 experiments (of 46) for which a cognitive model baseline is available. 8 experiments without a reported cognitive baseline are excluded to enable direct comparison across all columns. Finetuned families: Qwentaur, Llama-Centaur, Smoltaur, Olmotaur. Base columns: Q-8B/Q-14B (Qwen3), L-8B (Llama-3.1). Subscript _r denotes our reproducing evaluation of the original Centaur model distributed by [binz2025foundation], evaluated under identical python library and CUDA versions as our small foundation models for fair comparison. Subscript _p denotes values published by [binz2025foundation]. 70B_p is shown for reference but excluded from best/second-best marking, as it was evaluated under different software conditions. Cell colour encodes performance; **bold+underline** marks the best model, underline the second-best. Full per-experiment results in Appendix ??.

Task type	Qwentaur					Llama-Centaur			Smoltaur				Olmotaur		Base			[binz2025foundation]			Chance
	0.6B	1.7B	4B	8B	14B	1B	3B	8B	0.1B	0.4B	1.7B	3B	1B	7B	Q-8B	Q-14B	L-8B	70B _p	70B _r	Cog _p	ln(<i>k</i>)
Decision (9)	0.29	0.29	0.29	0.28	0.28	0.29	0.29	<u>0.28</u>	0.36	0.33	0.32	0.28	0.33	0.29	0.41	0.40	0.41	0.27	0.29	0.46	0.62
MDP (5)	0.54	0.53	0.52	0.52	0.51	0.54	0.53	0.52	0.62	0.60	0.56	0.53	0.61	0.53	0.64	0.62	0.61	0.51	<u>0.52</u>	0.69	0.77
Bandit (12)	0.60	0.60	0.59	<u>0.58</u>	0.58	0.62	0.60	0.58	0.73	0.68	0.64	0.59	0.66	0.60	0.74	0.74	0.76	0.60	0.61	0.75	1.15
Memory (4)	0.45	0.45	0.44	0.44	0.44	0.46	0.45	0.44	0.57	0.51	0.48	0.45	0.49	0.45	0.59	0.54	0.59	0.44	<u>0.44</u>	0.69	1.20
Misc. (2)	0.50	0.49	0.48	0.48	0.48	0.50	0.49	<u>0.48</u>	0.62	0.53	0.51	0.48	0.51	0.49	0.67	0.66	0.68	0.49	0.49	0.51	1.24
Sup. learn. (3)	0.75	0.76	0.74	0.74	0.74	0.76	0.74	<u>0.74</u>	0.93	0.87	0.84	0.74	0.93	0.74	0.82	0.81	0.83	0.74	0.74	1.05	1.19
Mean (35)	0.50	0.50	0.49	<u>0.49</u>	0.48	0.51	0.50	0.49	0.61	0.57	0.54	0.49	0.56	0.50	0.63	0.61	0.63	0.49	0.50	0.67	0.97
<0.20 0.20–0.40 0.40–0.60 0.60–0.80 0.80–1.00 >1.00																					

Table 5: Mean NLL by task type for finetuned models (bf16), restricted to the 35 experiments (of 46) that have both a reported cognitive model baseline and a well-defined discrete response space admitting a chance-level baseline $\ln(k)$, where k is the number of response options. 11 experiments lacking either a cognitive baseline or a computable $\ln(k)$ (continuous, mixed, or variable action spaces) are excluded. Finetuned families: **Qwentaur**, **Llama-Centaur**, **Smoltaur**, **Olmotaur**. Base columns: Q-8B/Q-14B (Qwen3), L-8B (Llama-3.1). Subscript _{*r*} denotes our reproducing evaluation of the original Centaur model distributed by [binz2025foundation], evaluated under identical python library and CUDA versions as our small foundation models for fair comparison. Subscript _{*p*} denotes values published by [binz2025foundation]. 70B_{*p*} is shown for reference but excluded from best/second-best marking, as it was evaluated under different software conditions. Cell colour encodes performance; **bold+underline** marks the best model, underline the second-best. Full per-experiment results in Appendix ??.

	Qwentaur					Llama-Centaur			Smoltaur				Olmotaur		Base			[binz2025foundation]		
Task type	0.6B	1.7B	4B	8B	14B	1B	3B	8B	0.1B	0.4B	1.7B	3B	1B	7B	Q-8B	Q-14B	L-8B	70B _p	70B _r	Cog _p
Decision (8)	0.53	0.53	0.54	0.54	<u>0.54</u>	0.53	0.53	<u>0.54</u>	0.41	0.46	0.48	0.54	0.47	0.53	0.34	0.35	0.33	0.56	0.52	0.27
MDP (5)	0.28	0.30	0.31	0.31	<u>0.32</u>	0.28	0.30	0.31	0.18	0.20	0.26	0.30	0.20	0.30	0.15	0.17	0.19	0.32	<u>0.31</u>	0.11
Bandit (12)	0.48	0.48	0.49	<u>0.49</u>	<u>0.50</u>	0.46	0.48	0.49	0.36	0.41	0.44	0.49	0.42	0.48	0.35	0.36	0.33	0.46	0.45	0.39
Memory (4)	0.57	0.57	0.58	0.58	<u>0.58</u>	0.56	0.57	0.58	0.46	0.51	0.55	0.57	0.53	0.57	0.42	0.47	0.43	0.58	<u>0.58</u>	0.36
Misc. (2)	0.57	0.57	0.58	0.58	<u>0.58</u>	0.56	0.57	<u>0.58</u>	0.45	0.53	0.55	0.58	0.55	0.57	0.41	0.42	0.40	0.57	0.57	0.55
Sup. learn. (3)	0.30	0.29	0.31	0.31	<u>0.31</u>	0.29	0.31	0.31	0.18	0.23	0.25	0.30	0.20	<u>0.31</u>	0.24	0.24	0.23	0.30	0.30	0.11
Mean (34)	0.46	0.46	0.47	0.48	<u>0.48</u>	0.45	0.47	<u>0.48</u>	0.35	0.39	0.43	0.47	0.40	0.47	0.32	0.34	0.32	0.47	0.46	0.30
		<0.05	0.05–0.12		0.12–0.18		0.18–0.25		0.25–0.32		0.32–0.40		0.40–0.48		0.48–0.55		≥0.55			

Table 6: Fraction of available information captured above chance, $(\ln k - \text{NLL}) / \ln k$, by task type for finetuned models (bf16). A value of 0 indicates chance-level performance; 1 indicates perfect prediction. Restricted to 34 experiments (of 46) with both a cognitive model baseline and a well-defined discrete response space ($\ln k > 0$); 12 experiments are excluded. Finetuned families: Qwentaur, Llama-Centaur, Smoltaur, Olmotaur. Base columns: Q-8B/Q-14B (Qwen3), L-8B (Llama-3.1). Subscript _r denotes our reproducing evaluation of the original Centaur model distributed by [binz2025foundation], evaluated under identical python library and CUDA versions as our small foundation models for fair comparison. Subscript _p denotes values published by [binz2025foundation]. 70B_p is shown for reference but excluded from best/second-best marking, as it was evaluated under different software conditions. **Bold+underline** marks the best model, underline the second-best. Full per-experiment results in Appendix ??.

Experiment	Type	Qwen 2.5/3 family						Llama 3/3.1/3.2 family						[binz2025foundation]						Chance																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
		Base			Non-cog.			Other cog.			Non-cog.			Other cog.			Non-cog.				Other cog.																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																			
		8B	14B	Qentaur	8B	14B	Socr.-14B	Herm.-14B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B		8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B	8B	3B

 <0.20
  0.20–0.40
  0.40–0.60
  0.60–0.80
  0.80–1.00
  1.00–1.20
  1.20–1.50
  1.50–2.00
  >2.00

Bold+underline = best; underline = second-best. Lower is better. "—" = not available. Models are grouped by base-model family. "Base" = pretrained model without finetuning (bf16); "Other cog." = other cognitive/behavioural foundation models finetuned on non-Psych-101 behavioural data; "Non-cog." = models finetuned for general instruction-following or reasoning. Herm. = Hermes [teknium2024hermes]; Memo. = Nemotron [nvdia2025nemotron]; Socr. = Socrates [kolluri2025finetuning]; Be-FW = [xie2025fn]. Base model versions differ within families: Qentaur uses Qwen3; Socr.-14B uses Qwen2.5; Herm.-14B uses Qwen2.5; Llama-Centaur uses Llama-3.1/3.2; Socr.-8B uses Llama-3; Be-FW uses Llama-3.1.

Table 7: Comparison of cognitively finetuned models with other behavioural foundation models and non-cognitive controls on Psych-101, grouped by base-model family. Within each family, **Qentaur**/Llama-Centaur (cognitive SFT on Psych-101) consistently outperform both other behavioural FMs and non-cognitive controls, demonstrating that alignment with human behavioural patterns requires cognitive finetuning on structured experimental data, not merely finetuning per se.